

Supplementary Table S1. FcμR residues that make intermolecular contacts (≤ 5 Å) with Fcμ chains across different complexes

PDB ID	FcμR chain	Contacted residues	Key interactions	Notes
7YTE	FcμR1	45, 57-60, 63, 65-67, 69, 107-111	salt bridges (45,111); H-bonds (60,63,110); hydrophobic (67)	Missing 42-44, 106, 112 (cf. 7YSG, 8BPG)
	FcμR2	45, 57-60, 63, 65-67, 69, 107-111	salt bridges (45,69,111); H-bonds (60,63,110); hydrophobic (67)	Retains residue 71 contact (same as 7YSG FcμR2)
8BPG	FcμR1	42-45, 57-60, 63-67, 69, 106-112	salt bridges (45,111); H-bonds (60,63,110); hydrophobic (67)	Identical to 7YTE FcμR1 except additional 42-44, 106, 112
	FcμR2	42-45, 57-60, 63-67, 69, 106-112	salt bridges (45,111); H-bonds (60,63,110); hydrophobic (67)	Same as 7YTE FcμR1
7YSG	FcμR1	44, 45, 57-60, 63-67, 108-112	salt bridges (45,111); H-bonds (60,63,110); hydrophobic (67)	–
	FcμR2	42-45, 57-60, 63, 65-67, 69, 106-112	salt bridges (45,69,111); H-bonds (60,63,110); hydrophobic (67)	Retains residue 71 contact (same as 7YTE FcμR2)
	FcμR3	42-45, 57-60, 63, 65-69, 106-112	salt bridges (45,111); H-bonds (60,63,110); hydrophobic (67)	Lacks Ser63; Phe67 contact verified
	FcμR4	42-45, 57-60, 65-69, 106-112	salt bridges (45,111); H-bonds (60,110); hydrophobic (67)	Same as FcμR3

All distances measured in PyMOL 3.1.6.1 between any side-chain or backbone heavy atom of FcμR-D1 (residues 18-124) and IgM partner chains. A contact is defined as at least one atom pair ≤ 5 Å. Continuous residue ranges indicate sequential coverage. Key interaction types (salt bridges, hydrogen bonds, hydrophobic packing) were assigned using PLIP and PDBePISA.

Supplementary Table S2. Backbone-superposition RMSD of FcμR-D1 alanine mutants onto wild-type chain C (PDB 7YTE)

FcμR mutant	RMSD value (Å)	Match Align score	Initial aligned atoms	Refined atoms	Rejected atoms
R45A	0.000	547	791	791	0
T57A	0.000	543	791	791	0
T60A	0.000	543	795	791	4
S63A	0.000	545	796	792	4
T65A	0.000	543	791	791	0
N66A	0.000	540	790	790	0
F67A	0.000	549	787	787	0
K69A	0.000	542	793	789	4
T110A	0.000	548	797	797	0
D111A	0.000	540	794	790	4
R112A	0.000	542	791	787	4
T60A/S63A	0.000	540	794	786	8
T110A/D111A	0.000	535	792	784	8
T60A/S63A/D111A	0.000	532	791	779	12
R45A_F67A_K69A	0.000	522	775	759	16
R45A_F67A_R112A	0.000	528	779	767	12

Superpositions were performed in PyMOL 3.1.6.1 using the align command with chain C as reference. Rejected atoms correspond to mutated side-chain atoms excluded during iterative RMSD pruning.

Superpositions were performed in PyMOL 3.1.6.1 using the align command with chain C as the reference. The difference between “Initial aligned atoms” and “Refined atoms” (0–16 atoms per mutant) consists exclusively of side-chain atoms at the mutated positions, which were rejected during a single pruning cycle. All refined atom pairs yielded an RMSD of 0.000 Å, confirming that none of the alanine substitutions caused backbone conformational changes.

Supplementary Table S3. Software and web resources used in this study

Software / Resource	Version	Purpose	Key parameters
MDAnalysis	2.9.8	C α RMSD calculations	Python 3.12
UCSF ChimeraX	1.10	Structural superposition, hydrogen bond analysis	MatchMaker: Needleman-Wunsch, BLOSUM-62
PyMOL	3.1.6.1	Alanine mutagenesis, structure superposition, visualization, RMSD calculation	Align command, default outlier pruning
Biopython	1.85	Inter-residue contact analysis	Side-chain distance cutoff 4.5 Å
PLIP	2.3.0	Protein–protein interaction profiling	Accessed via https://plip-tool.biotec.tu-dresden.de/plip-web/plip/index , default settings
PDBePISA	–	Interaction validation	Accessed via https://www.ebi.ac.uk/msd-srv/prot_int/cgi-bin/piserver , default settings
PyMOL	3.1.6.1	Atomic visualization, alanine points mutate, RMSD calculation	Distance cutoff 3.0 Å; Mutagenesis Wizard
ConSurf	–	Evolutionary conservation analysis	UniRef90 database, empirical Bayesian inference
MAFFT	7	Multiple sequence alignment	Accessed via https://mafft.cbrc.jp/alignment/server/ , default settings
FoldX	5.0	Alanine scanning, $\Delta\Delta G$ prediction	0 independent runs, 298 K, pH 7.0, ionic strength 0.05 M, water = CRYSTAL, vdwDesign = 2
PremPS	–	Alanine scanning, $\Delta\Delta G$ prediction	Accessed via https://lilab.jysw.suda.edu.cn/research/PremPS/ , default settings

–, version indicates not formally released; the latest available between 2025 and 2026 was used. All web-based tools were accessed with default parameters unless otherwise specified in Methods.